

Risk Management in Energy Projects Using Fuel Price Forecasting

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Abstract—Energy projects involve substantial capital investments whose financial viability depends on accurate cost estimation and effective management of operational expenses. Fuel price volatility, driven by macroeconomic conditions, geopolitical developments, and supply–demand imbalances, poses significant financial and cost-related risks that threaten project profitability. Reactive responses to fuel price fluctuations often result in budget overruns and inefficient corrective actions.

This study examines how fuel price forecasting can be integrated into energy project risk management frameworks to support proactive and informed decision-making. An analytical and descriptive approach is employed, synthesizing existing literature on fuel price volatility and forecasting methodologies alongside an analysis of historical Brent crude oil price data obtained from the Federal Reserve Economic Data system and the World Bank Commodity Price Database. Rather than developing forecasting models, the research focuses on translating forecasting insights into practical managerial actions, including contingency planning, budget development, procurement strategies, and cost control.

The study demonstrates that integrating fuel price forecasting into decision processes enables project managers to move beyond static cost assumptions and develop structured, risk-informed responses aligned with market conditions, thereby enhancing project resilience, profitability, and stakeholder confidence.

Index Terms—Energy projects, fuel price volatility, risk management, forecasting, machine learning

I. INTRODUCTION

Energy projects are among the most capital-intensive investments worldwide, involving large upfront costs and long operational lifespans. The financial viability of such projects depends heavily on accurate cost estimation and effective management of operational expenses. Among the various cost drivers influencing project performance, fuel costs play a critical role, particularly in energy projects that rely on fossil fuels. As a major recurring operational expense, fuel prices have a direct and substantial impact on project profitability, making fuel price volatility a key source of financial risk that must be addressed through structured risk management practices.

Fuel prices, especially crude oil prices that serve as a global benchmark for energy markets, are inherently volatile. This volatility arises from a combination of macroeconomic conditions, geopolitical developments, supply–demand imbalances, and financial market dynamics. Sudden and significant price

fluctuations can undermine project cost estimates, disrupt budget planning, and increase uncertainty in investment decision-making. Historical oil price movements demonstrate that unanticipated price changes have rendered otherwise viable projects financially unsustainable, highlighting the importance of proactive approaches to managing fuel price risk.

Effective risk management enables project organizations to systematically identify, assess, and respond to risks that threaten project objectives. Rather than reacting to fuel price changes after they occur, proactive risk management seeks to anticipate potential price movements and prepare appropriate mitigation strategies in advance. Fuel price forecasting supports this approach by providing insights into potential future price ranges and volatility patterns based on historical data and market behavior. By integrating forecasting outputs into managerial decisions such as contingency reserve allocation, budget planning, and procurement strategy project managers can reduce exposure to adverse price movements and enhance financial resilience. Accordingly, this research examines how fuel price forecasting can be incorporated into energy project risk management frameworks to support improved decision-making and more effective control of fuel price related risks.

II. PROBLEM STATEMENT

Energy projects depend critically on accurate cost estimation and effective management of operational expenses throughout their lifecycle. However, a fundamental challenge in energy project planning is the pervasive reliance on static fuel price assumptions that fail to account for the substantial volatility and uncertainty inherent in global fuel markets. Fuel price volatility, driven by macroeconomic conditions, geopolitical events, and supply–demand imbalances, creates significant financial and cost-related risks that project budgets and planning processes are often inadequately prepared to address. This inadequate consideration of fuel price uncertainty results in recurring budget overruns, difficulties in cost control during execution, and reactive management approaches that address price changes only after they have already impacted project performance. The problem is not the existence of fuel price volatility itself—which is an unavoidable characteristic of energy markets—but rather the absence of systematic, proactive risk management approaches that anticipate likely price movements and integrate them into project planning and execution

decisions. While forecasting methodologies exist in academic literature and professional practice, significant gaps remain in translating these forecasting capabilities into actionable intelligence that systematically informs project management decisions regarding contingency planning, budgeting, procurement strategy, and cost control. This research addresses the gap between the technical capability to forecast fuel prices and the practical integration of forecasting insights into energy project risk management frameworks.

III. LITERATURE REVIEW

A. Fuel Price Volatility and Risk in Energy Projects

Volatility of fuel prices represents one of the most significant sources of uncertainty in energy project management, directly affecting cost structures, financial performance, and investment decision-making. The fluctuation of crude oil prices creates substantial challenges for project budgeting and long-term financial planning, as energy projects typically involve large capital investments with extended operational lifespans. These price movements introduce both financial risk and market risk, potentially affecting project viability and profitability [1].

The impact of oil price volatility extends beyond simple cost estimation errors. As demonstrated in studies on crude oil price volatility and value-at-risk, price fluctuations can significantly affect project cash flows, return on investment calculations, and the overall risk profile of energy ventures [1]. This volatility complicates the investment evaluation process, as traditional financial metrics such as net present value and internal rate of return become highly sensitive to fuel price assumptions. Consequently, project managers and investors face substantial uncertainty when evaluating the economic feasibility of energy projects, particularly in markets characterized by high price instability [1].

Furthermore, volatility in fuel prices directly influences procurement strategies, operational budgets, and emergency planning requirements. The unpredictable nature of oil markets means that energy projects must account for potential cost overruns arising from fuel price escalation, which can erode profit margins and jeopardize project completion within budget constraints [1]. This underscores the critical need for robust risk management frameworks that can anticipate and mitigate the adverse effects of fuel price uncertainty on project outcomes.

B. Fuel Price Forecasting Approaches

Forecasting fuel prices has emerged as a vital tool for understanding and managing the inherent uncertainty in energy markets. The academic literature presents a diverse range of forecasting methodologies, broadly categorized into traditional econometric approaches and advanced machine learning techniques. These methods aim to capture the complex dynamics of fuel price movements and provide decision-makers with insights into future price trends and volatility patterns.

Traditional econometric models, particularly Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models, have been extensively employed for fuel price forecasting. ARIMA models capture temporal dependencies and trends in price series, while GARCH models specifically address volatility clustering and time-varying variance characteristics observed in oil markets [2]. These approaches have demonstrated reasonable performance in capturing short-term price dynamics and volatility patterns, providing a foundation for understanding fuel price behavior.

However, the complexity and nonlinearity inherent in oil price movements have motivated researchers to explore advanced forecasting techniques. Hybrid machine learning models, which combine econometric foundations with artificial intelligence capabilities, have shown promise in improving forecast accuracy and capturing nonlinear relationships in fuel price data [2]. These hybrid approaches leverage the strengths of both traditional statistical methods and modern computational techniques to enhance predictive performance [3].

In the era of big data, forecasting methodologies have evolved to incorporate larger datasets and more sophisticated analytical frameworks. Research has demonstrated that machine learning techniques can effectively process diverse information sources and identify complex patterns that traditional models may overlook [3]. The integration of big data analytics into fuel price forecasting represents a significant advancement in understanding market dynamics and volatility drivers.

Energy market volatility modeling has also benefited from the combination of GARCH frameworks with machine learning algorithms. This integration allows for more nuanced capture of volatility persistence, asymmetric effects, and regime changes that characterize energy markets [5]. By combining econometric rigor with computational flexibility, these hybrid approaches provide more comprehensive tools for forecasting fuel price volatility and understanding market uncertainty [5].

C. Linking Fuel Price Forecasting to Risk Management

While forecasting methodologies provide valuable insights into future price movements and volatility, their practical utility depends on effective integration into risk management decision-making processes. Fuel price forecasts can support several critical risk management functions in energy project management, including scenario analysis, contingency planning, budgeting, and procurement strategy development.

Scenario analysis represents one of the most direct applications of fuel price forecasting in project risk management. By generating multiple price trajectories based on different assumptions and models, project managers can evaluate project performance under various market conditions and identify vulnerabilities to price shocks. This enables more informed decision-making regarding project design, timing, and risk mitigation strategies.

Contingency reserve planning also benefits significantly from fuel price forecasting. Accurate volatility predictions

allow project managers to establish appropriate contingency buffers that account for potential fuel cost escalations without unnecessarily tying up capital in excessive reserves. This optimization of contingency planning contributes to more efficient resource allocation and improved project financial performance.

In terms of budgeting and cost control, fuel price forecasts provide a foundation for developing realistic cost estimates and identifying potential budget pressure points throughout the project lifecycle. This forward-looking perspective enables proactive adjustments to project plans and procurement strategies before cost overruns materialize.

The application of artificial intelligence to market risk prediction in oil markets has demonstrated the potential for advanced forecasting techniques to support risk management decision-making [4]. Machine learning models can identify complex patterns and relationships that inform market risk assessment and help stakeholders anticipate potential price movements [4]. However, existing research in this domain has primarily focused on market-level risk prediction rather than the integration of forecasting insights into project-level risk management frameworks [4].

Research Gap

While extensive research exists on fuel price volatility characteristics and forecasting methodologies, limited attention has been given to the systematic integration

of forecasting insights into energy project risk management decision-making frameworks. The existing literature has largely treated forecasting as an analytical exercise focused on predictive accuracy, with less emphasis on how forecast outputs can be operationalized within project management practices. Specifically, there is a gap in understanding how fuel price forecasting can be embedded into structured risk management processes such as risk identification, assessment, response planning, and monitoring in the context of energy projects.

Furthermore, most studies on market risk prediction using forecasting techniques have concentrated on financial market applications and trading strategies rather than project management contexts [4]. This leaves a significant knowledge gap regarding the practical implementation of forecasting tools in energy project planning, budgeting, and risk mitigation. The current study addresses this gap by developing a framework that explicitly links fuel price forecasting outputs to project management risk management practices, demonstrating how volatility predictions and price scenarios can inform contingency planning, budget development, and procurement decisions in energy projects. This integration provides a bridge between forecasting analytics and actionable risk management strategies, contributing to more resilient and financially robust energy project execution.

IV. Research Methodology

IV.A Research Design

This study adopts an applied analytical research design that integrates time-series forecasting with project risk management principles. A practical forecasting experiment is conducted using historical Brent crude oil price data to generate short-term price predictions. The objective is not to achieve maximum predictive accuracy, but to produce realistic price estimates that can support scenario-based risk planning. By linking statistical forecasting outputs with project management decision-making processes, the study bridges the gap between analytical modeling and practical risk control in energy projects.

IV.B Data Sources

Historical Brent crude oil price data were used as a proxy for fuel cost dynamics in energy projects. The dataset was obtained from the Federal Reserve Economic Data (FRED) database, which provides publicly accessible and reliable macroeconomic time-series data. The original data were available at daily frequency and were later aggregated into monthly averages to reduce short-term noise and capture broader market trends. The selected time period spans multiple market cycles, including stable periods and episodes of high volatility, allowing for a more realistic forecasting exercise.

IV.C Data Preparation

Before model estimation, the dataset was examined for completeness and structural consistency. The time index was converted into a proper date format and set as the primary time reference for the series. Since the original data were daily, the series was resampled into monthly averages to ensure smoother trend representation and reduce high-frequency volatility.

To evaluate forecasting performance objectively, the dataset was divided into two subsets: a training set and a testing (holdout) set. The majority of historical observations were used for model training, while the most recent 12 months were reserved for validation. This separation prevents data leakage and enables an unbiased assessment of model performance.

IV.D Forecasting Approach

To generate short-term price forecasts, an Autoregressive Integrated Moving Average (ARIMA) model was applied to the monthly Brent price series. ARIMA models are widely used in time-series forecasting due to their ability to capture trend behavior and temporal dependencies in economic data. A simple ARIMA (1,1,1) configuration was selected to balance model interpretability and forecasting performance.

To evaluate model reliability, the most recent 12 months of data were reserved as a holdout validation set. The model was trained on the remaining historical observations and then used to forecast the withheld period. Forecast accuracy was assessed using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). The results showed an MAE of approximately 7.79

USD and a MAPE of approximately 11.79%, indicating acceptable short-term predictive performance for scenario-based risk planning.

After validation, the model was re-estimated using the full dataset and used to generate a six-month forward forecast. These forecasted values serve as quantitative inputs for constructing structured risk scenarios in the subsequent analysis

V. Forecasting and Scenario Analysis

Model Validation

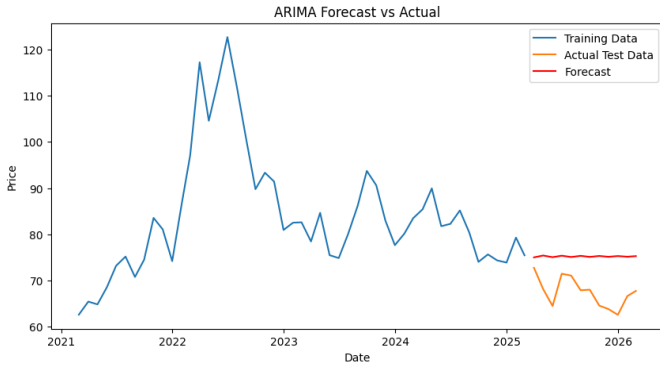


Figure 1: ARIMA model validation: forecasted values compared with actual test data

V.1 Forecast Output

Using the calibrated ARIMA model trained on the full monthly dataset, a six-month forward forecast was generated to estimate short-term Brent crude oil price behavior. The base forecast indicates a relatively stable price level around approximately 67 USD per barrel over the upcoming months. While minor fluctuations are observed, the general projection suggests short-term price stabilization within a narrow range.

This forecast represents the statistical best estimate derived from historical trend dynamics and serves as the baseline input for subsequent scenario construction and risk evaluation

V.2 Scenario Construction

To translate the statistical forecast into actionable risk management inputs, three structured price scenarios were constructed. The **base scenario** corresponds directly to the ARIMA point forecast. A **high-price scenario (+15%)** was introduced to simulate adverse market conditions reflecting potential upward volatility. Similarly, a **low-price scenario (-10%)** was defined to represent favorable short-term market movement.

These percentage adjustments do not represent additional predictive claims; rather, they function as structured stress-testing assumptions. By creating a bounded price range around the baseline forecast, project managers can evaluate budget sensitivity and contingency adequacy under plausible market fluctuations.

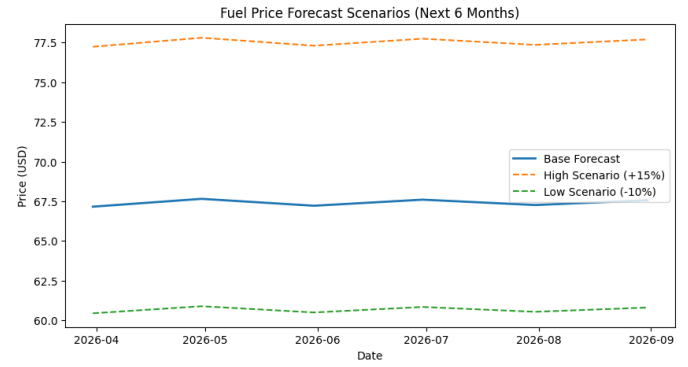


Figure 2: Six-month forecast scenarios including base, high (+15%), and low (-10%) projections

V.3 Managerial Interpretation

The constructed scenarios provide a structured framework for translating fuel price uncertainty into project-level financial implications. Rather than relying on a single static cost assumption, the scenario range allows project managers to assess potential deviations from the baseline budget.

Under the high-price scenario, fuel-related operational costs would increase proportionally, potentially requiring larger contingency reserves, tighter cost monitoring, or revised procurement strategies. Conversely, the low-price scenario reflects favorable market conditions that may reduce cost pressure and improve budget flexibility.

By integrating forecast-based scenarios into planning processes, project managers can shift from reactive cost adjustments to proactive risk mitigation. This approach strengthens budget resilience, improves decision transparency, and enhances overall project financial stability in volatile energy markets

VI. Risk Management Application

The forecast-based scenarios developed in Section V provide a quantitative foundation for translating fuel price uncertainty into structured risk management interventions aligned with established project management methodologies. Drawing on risk-based reserve allocation principles [6] and scenario analysis frameworks documented in energy infrastructure planning literature [7, 10], this section demonstrates the operational application of forecasting outputs across four critical decision domains: contingency reserve determination, procurement strategy formulation, budget development under uncertainty, and performance monitoring systems. The integration of these applications constitutes a comprehensive risk management architecture enabling project organizations to transition from reactive cost adjustments to proactive, data-informed decision protocols.

VI.A Contingency Reserve Allocation

Contingency reserve allocation represents a critical mechanism for absorbing cost variance arising from identified risks without requiring emergency funding approvals or scope reductions. The scenario range derived from ARIMA forecasting enables quantitative contingency determination linked directly to market uncertainty. This approach aligns with risk-based contingency planning methodologies documented in project management literature [6], wherein reserve sizing derives from quantified risk impact assessments rather than arbitrary percentage allocations.

For a hypothetical project consuming 10,000 barrels of fuel monthly, the base scenario implies a monthly fuel cost of approximately 670,000 USD. The high-price scenario increases this to 770,000 USD—a difference of 100,000 USD per month. Over a six-month period, this differential totals 600,000 USD. This quantification provides an objective basis for determining whether the existing contingency reserve is adequate or requires augmentation. If the current reserve falls below this threshold, project managers have early warning to secure additional funding or implement cost mitigation measures before price increases materialize.

VI.B Procurement Strategy Optimization

Procurement timing and contract structure represent critical levers for managing fuel price exposure. When the forecast indicates short-term price stability (as demonstrated by the relatively flat base scenario projection), project managers may adopt spot market purchasing strategies to maintain procurement flexibility and avoid locking in prices during periods of expected stability. Conversely, if forecasting models indicate rising price trends or increasing volatility, forward contracts or hedging instruments become more attractive despite higher upfront costs or reduced flexibility.

The scenario range also informs contract negotiation parameters. If the forecast suggests potential upward price movement (reflected in the high-price scenario), procurement managers can negotiate price ceiling clauses or escalation limits in supply contracts. These contractual protections cap the project's exposure to adverse price changes while preserving supplier relationships. Additionally, the low-price scenario provides insight into favorable procurement windows where bulk purchasing or inventory accumulation may be cost-effective, provided that storage costs and working capital constraints permit such strategies.

VI.C Scenario-Based Budget Development

Conventional budgeting methodologies typically rely on deterministic fuel price assumptions that rapidly become obsolete in volatile commodity markets. Scenario-based budgeting replaces this static approach with a range-based framework that explicitly acknowledges price uncertainty, aligning with scenario planning principles documented in strategic management and energy planning literature [7, 8, 9]. Rather than presenting a single cost estimate vulnerable to immediate obsolescence, scenario-based budgets define a price envelope encompassing plausible market outcomes.

This approach enables more realistic budget presentations to

stakeholders. Rather than providing a single cost estimate that may be exceeded if prices rise, project managers can present a range that reflects market uncertainty. For example, the project budget might specify: "Fuel costs are projected at 670,000 USD monthly under stable market conditions (base scenario), with potential variation between 603,000 USD (low scenario) and 770,000 USD (high scenario) depending on market developments." This transparency improves stakeholder confidence by demonstrating that price risk has been systematically assessed rather than ignored.

VI.D Cost Control and Performance Monitoring

Effective cost control requires continuous monitoring of actual fuel costs against forecasted scenarios and predefined variance thresholds. The forecast scenarios establish quantitative benchmarks for evaluating project performance. When actual fuel prices exceed the high-price scenario or fall below the low-price scenario, these deviations trigger management reviews to determine whether corrective actions are necessary.

Variance analysis between forecasted and realized prices also provides valuable feedback for improving future forecasting accuracy. If systematic forecast errors are detected—such as consistent underestimation of price increases—project managers can adjust their forecasting approach or scenario construction methodology to better reflect actual market behavior. This creates a continuous improvement cycle where forecasting performance is evaluated and refined based on observed outcomes, enhancing the effectiveness of risk management over time.

Furthermore, regular forecast updates allow project managers to adapt to changing market conditions. As new data becomes available and market dynamics evolve, the forecasting model can be re-estimated to generate updated scenarios. This iterative approach ensures that risk management decisions remain aligned with current market expectations rather than relying on outdated projections that no longer reflect prevailing conditions.

VII. Discussion

The integration of time-series forecasting methodologies into energy project risk management frameworks demonstrates that systematic translation of statistical predictions into operational decisions can substantially enhance risk mitigation effectiveness. While ARIMA-based forecasting constitutes an established econometric technique [2], the principal contribution of this research lies in developing a replicable framework for converting forecast outputs into concrete management interventions across four critical decision domains. This framework addresses a documented gap in existing literature, which frequently emphasizes forecasting accuracy while providing limited guidance on operationalizing forecast results within project management processes [7, 9].

The primary contribution of this study lies not in the forecasting model itself—ARIMA models are well-established in time-series analysis [2, 3]—but rather in the systematic framework for translating forecast outputs into practical risk management interventions. The scenario-based approach bridges the gap between statistical predictions and operational decision-making by converting probabilistic price ranges into concrete financial implications that project managers can act upon. This translation

process addresses a critical limitation in existing literature, which often focuses on forecasting accuracy while providing limited guidance on how forecast results should inform project-level decisions [7, 9].

VII.A Methodological Considerations and Limitations

Several limitations must be acknowledged when interpreting these results. First, the ARIMA model employed in this study assumes that future price behavior follows patterns observed in historical data. This assumption may not hold during periods of structural market change, such as the introduction of new energy technologies, major geopolitical disruptions, or fundamental shifts in global energy policy. During such periods, historical patterns may lose predictive validity, and forecast accuracy may decline accordingly.

Second, the scenario construction approach relies on percentage adjustments (+15% and -10%) that, while grounded in observed market volatility, remain somewhat arbitrary. Alternative scenario definitions could yield different risk assessments. Future research might explore more sophisticated scenario generation methods that incorporate volatility modeling or extreme value analysis to better characterize tail risks—low-probability, high-impact events that could severely affect project finances.

Third, this study uses Brent crude oil as a proxy for fuel costs, which is appropriate for many energy projects but may not fully represent fuel cost dynamics in projects using natural gas, coal, or renewable energy sources. Each fuel type exhibits distinct price volatility patterns and responds differently to market drivers. Project managers must adapt the forecasting methodology to the specific fuel types relevant to their projects.

VII.B Practical Implications for Project Management

The practical value of this approach extends beyond mere forecasting accuracy to encompass improved organizational decision-making processes. By systematically integrating fuel price forecasts into risk management frameworks, project organizations can enhance transparency in budget development, strengthen stakeholder communication, and establish clearer accountability for cost performance. When project managers present scenario-based budgets that explicitly acknowledge price uncertainty, stakeholders gain a more realistic understanding of project risks and can make better-informed decisions regarding project approval, funding levels, and performance expectations.

Moreover, the framework encourages a shift from reactive to proactive risk management culture. Traditional approaches often treat fuel price increases as unforeseen events requiring emergency responses. In contrast, forecast-informed risk management normalizes price volatility as an expected characteristic of energy projects and establishes predefined response protocols. This cultural shift reduces the likelihood of crisis-driven decision-making and enables more measured, strategic responses to market changes.

VII.C Directions for Future Research

Several avenues for future research emerge from this study. First, comparative analysis of different forecasting methodologies would provide valuable insights into which approaches offer the best balance between accuracy, computational complexity, and ease of interpretation for project management applications. While ARIMA models offer

transparency and interpretability, more sophisticated machine learning approaches such as ensemble methods or neural networks may improve forecasting accuracy, particularly during periods of high market volatility. Empirical comparison of these approaches using multiple fuel types and market conditions would help project managers select appropriate forecasting tools for their specific contexts.

Second, empirical validation through case studies would strengthen the practical applicability of this framework. Implementing the proposed approach in real energy projects and tracking outcomes over time would provide evidence regarding its effectiveness in reducing cost overruns, improving budget accuracy, and enhancing project performance. Such studies could also identify implementation challenges and organizational factors that facilitate or hinder successful integration of forecasting into risk management processes.

Finally, integration of fuel price forecasting with other project risk factors would enable more comprehensive risk management. Energy projects face multiple sources of uncertainty beyond fuel prices, including regulatory changes, technological developments, and macroeconomic conditions. Developing integrated risk models that simultaneously consider fuel price volatility alongside these other risk factors would provide project managers with a more complete understanding of overall project risk exposure and enable more coordinated mitigation strategies.

VIII. Conclusion

Integrating fuel price forecasting into risk management enhances budgeting accuracy, contingency planning, and financial resilience in energy projects. Bridging forecasting

analytics and project management decision-making supports more robust and proactive responses to fuel price uncertainty.

This research demonstrates that effective risk management in energy projects requires systematic integration of fuel price forecasting into operational decision-making frameworks rather than mere recognition of market volatility as an abstract concern. Through the application of ARIMA econometric modeling to historical Brent crude oil price data and subsequent construction of base, high-price, and low-price scenarios, this study establishes a methodologically rigorous yet operationally practical framework for translating statistical forecasts into actionable risk management interventions. The framework addresses four critical decision domains—contingency reserve allocation [6], procurement strategy optimization, scenario-based budget development [7, 8, 9], and cost control mechanisms—each contributing essential components to a comprehensive risk management architecture that enhances project financial stability and stakeholder confidence.

The significance of this contribution extends beyond the specific forecasting methodology employed. While ARIMA models constitute well-established econometric techniques with extensive validation in commodity price forecasting applications [2, 3], the research makes its primary contribution through systematic operationalization of forecast outputs within project management decision protocols. The scenario-based approach provides a replicable framework that project managers can adapt to diverse organizational contexts, fuel market conditions, and project-specific risk profiles, enabling implementation across varying technical capabilities, data availability constraints, and organizational risk tolerance levels.

The broader implications for energy project management practice are substantial. As energy markets undergo fundamental structural transformation driven by decarbonization policies, renewable energy integration, evolving regulatory frameworks, and shifting geopolitical dynamics, the complexity and uncertainty confronting project organizations will intensify rather than diminish. In this environment of accelerating change and persistent volatility, proactive risk management transitions from beneficial practice to organizational necessity. The integration of fuel price forecasting into risk management frameworks represents a fundamental paradigm shift from reactive crisis management to anticipatory strategic planning—a transition that will prove increasingly critical as stakeholders demand greater accountability for cost performance and financial resilience in an era of unprecedented market uncertainty.

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